

Polarization Noise Anisotropy-Aware Technique for Enhanced Earthquake Detection Using Coherent Subsea Transponders

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Abstract Polarization noise anisotropy, in coherent subsea transponders, limits the reliability of earthquake detection. We analyze lab- and field-deployed transponders using a tensor-based eigenvalue decomposition technique that dynamically aligns seismic signals with the eigenvectors, enhancing detection sensitivity up to 16 dB improvement in SNR. ©2026 The Author(s)

Introduction

Optical sensing using commercial fibers is a fast advancing field, increasingly established in Earth observation while being strategic and synergetic to optical communications. While technologies such as distributed acoustic sensing (DAS) are among the most mature technologies, polarization data from deployed coherent transponders in subsea routes have been proven to provide quality observations at negligible cost (no dedicated fiber and interrogator) or impact to their main telecom intent. Modern transponders possess sophisticated digital signal processing (DSP) capabilities, where internal polarization-tracking data can be redirected to serve as a distributed sensor. By continuously extracting the link's Jones matrix from adaptive equalizer coefficients [1], the system can detect external environmental perturbations such as earthquakes, pressure, or temperature changes. This transforms the existing communication infrastructure into a dual-purpose network that provides high-sensitivity sensing as a secondary benefit. Recent research has demonstrated the potential of tracking Jones rotation vectors for ultra-long-haul sensing, exemplified by the detection of Kamchatka earthquake precursors from 9,000 km away in the Mediterranean Sea [2]. In these systems, while the input state of polarization (SOP) remains stable for days, the received SOP fluctuates dynamically due to cumulative external perturbations along the fiber path. Despite the growing use of coherent transponders for seismic sensing, a fundamental challenge remains: polarization noise in optical fiber links is inherently anisotropic, meaning its intensity varies across different directions in the Stokes space [3]. Conventional detection approaches project seismic signals onto arbitrary fixed Stokes axes (S_1, S_2, S_3), which fails to account for this directional noise structure. As a result, background noise leaks into the measurement, artificially degrading the signal-to-noise ratio (SNR) and reducing the reliability of

earthquake detection. Furthermore, the relative contributions of terminal-side impairments and distributed fiber-induced noise, along with their spectral characteristics and anisotropy, remain insufficiently characterized. This work addresses these gaps by introducing an anisotropy-aware eigenvalue decomposition technique that dynamically aligns seismic signals with the lowest-noise axis, maximizing detection sensitivity.

Methodology

Environmental perturbations, such as hydrostatic pressure and axial strain, amplify the cylindrical asymmetry of fiber optic links, changing local birefringence. This physical evolution is governed by the Jones matrix, J , a 2×2 complex unitary matrix. Following the methodology in [4], we extract the three-dimensional rotation vector in Stokes space,

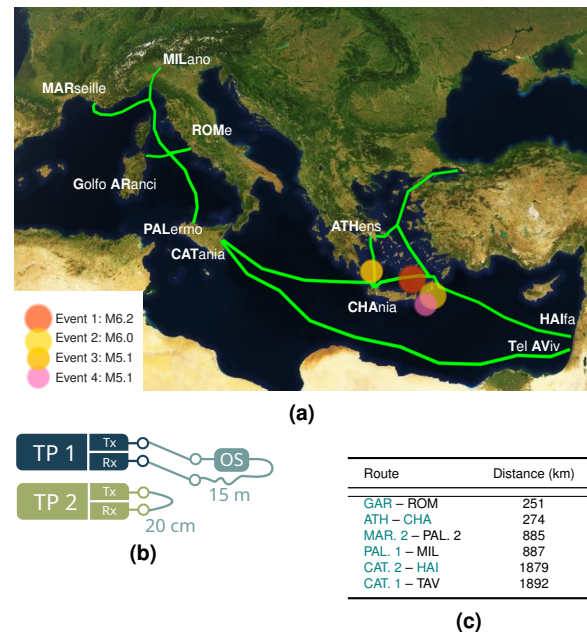


Fig. 1: (a) Geographical location of the cables, landing sites, and seismic events studied. (b) Lab testbed: each transponder studied in back-to-back and through ceiling cabling to and from an optical switch. (c) links between transponders in the Sparkle live networks.

$\phi(t) \in \mathbb{R}^3$, from J . The use of a slowly varying reference frame [5] isolates high-frequency environmental signals (e.g., acoustic-induced strain) from very slow polarization drifts, providing a complete, higher-dimensional representation of the birefringence state. At the coherent receiver, the cumulative birefringence manifests as a stochastic rotation of the output SOP characterized by J . Sensing detection limit can be described by its SNR, with the noise floor governed by terminal-side impairments (e.g., DSP algorithm, transponder hardware) and distributed environmental fluctuations. In long-haul submarine cables, distributed noise accumulates as a random walk of the rotation vector $\phi(t)$, possibly yielding a stochastically isotropic SOP distribution over time. To characterize the spatial dynamics of $\phi(t)$ independently of arbitrary geometric axes (S_1, S_2, S_3), we can compute its covariance tensor over a defined time window, analogous to the 3D polarization coherency matrix [6]:

$$T = \langle \phi(t) \phi(t)^\top \rangle \quad (1)$$

where $\langle \cdot \rangle$ denotes time averaging and $\phi(t)$ is a 3D column vector. Eigenvalue decomposition yields $T = Q\Lambda Q^\top$, with $\Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$ containing the eigenvalues ($\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$) that represent the signal variance along the principal axes. The degree of polarization anisotropy is quantified by the frame-invariant scalar metric A :

$$A = \frac{(\lambda_1 - \lambda_2)^2 + (\lambda_2 - \lambda_3)^2 + (\lambda_3 - \lambda_1)^2}{2(\lambda_1 + \lambda_2 + \lambda_3)^2} \quad (2)$$

where $A = 0$ indicates perfectly isotropic (random) directions, and $A \rightarrow 1$ denotes strictly anisotropic, single-axis behavior. While the global tensor T effectively characterizes steady-state background anisotropy, it geometrically partitions the energy of localized transient events (e.g., earthquakes) and the noise floor across arbitrary axes. This allows

the background noise to leak into the measurement, artificially degrading the actual SNR. To optimally isolate coordinate bias, we compute a rolling-window covariance tensor T_W over window W . By dynamically projecting the raw time-domain rotation vector onto the instantaneous primary eigenvectors $v_i(t)$ of T_W using $\phi_{\lambda_i}(t) = \phi^\top v_i(t)$, the signal energy is realigned with the correct and consistent axis. Standard spectral estimators, such as Welch's method, can then be applied independently to each decoupled component, yielding a directional power spectral density (PSD) that truly reflects the noise structure along each principal axis.

Results

In this study, we analyze data collected from a total of ten transponders [7], including two in laboratory links and eight deployed across operational Sparkle links, as illustrated in Fig. 1. The two laboratory transponders (Fig 1b) were tested under two distinct lab configurations, back-to-back connected with i) short length of fiber about 20 cm (referring as S) and ii) through ceiling cabling to and from an optical switch which is longer around 20 m (referring as L) to assess the impact of controlled conditions with two different transponders, while the field-deployed units capture real environmental and network-induced perturbations. The routes and the geodesic distances are also shown in Fig 1c. For each transponder, Jones matrix measurements were recorded continuously at an approximate sampling rate of 2 Hz. From these measurements, the corresponding rotation vectors $\phi(t)$ were subsequently extracted. Although all transponders are based on the same hardware architectures, minor variations in operational conditions – of the outputs of the DSP engine – are expected as their capacity, i.e., the transmission

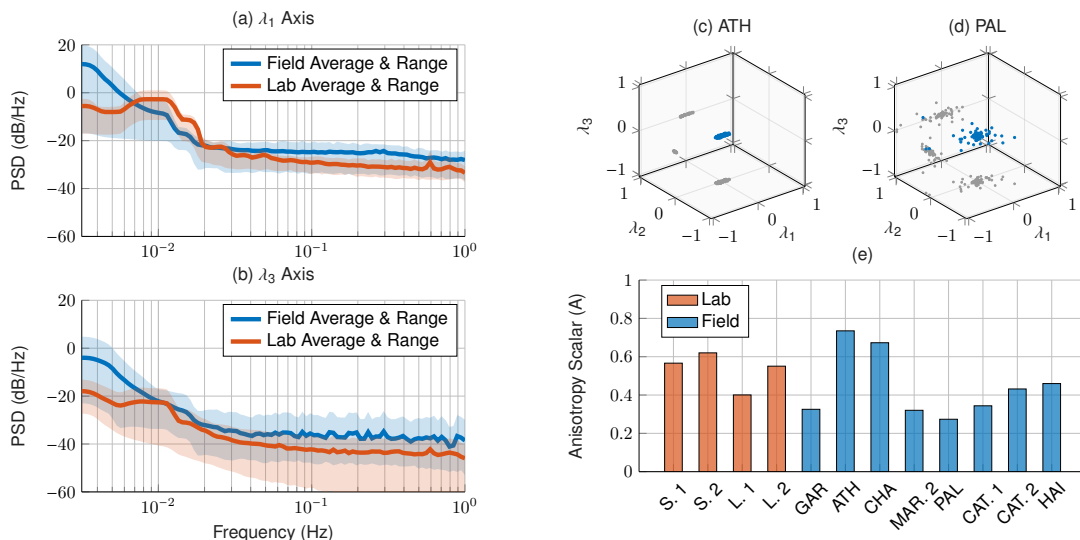


Fig. 2: The average PSD along the (a) λ_1 and (b) λ_3 axis. (c-d) Visualization of the rotation vector for two field transponders (decomposed every 15 minutes) and (e) the average anisotropy metric of all studied data sets.

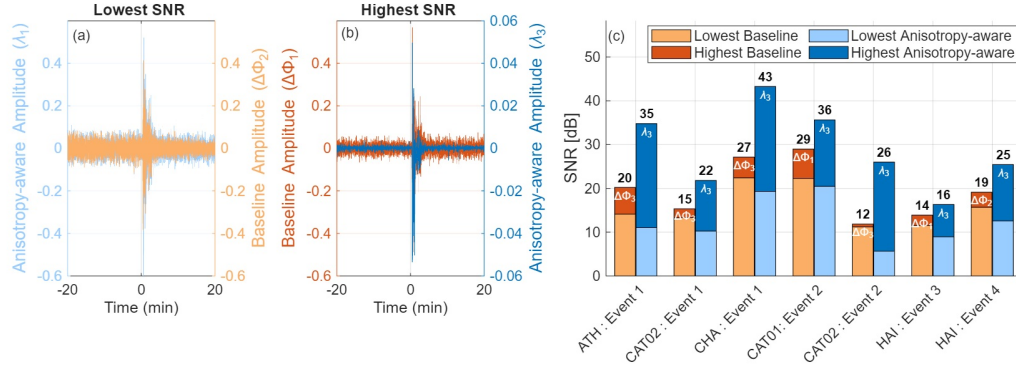


Fig. 3: Amplitude on different axes resulting (a) the lowest and (b) the highest SNR for baseline and anisotropy-aware approach for transponder CAT. 1 and Event 2, and (c) the best peak SNR enhancement for various pairs of transponder-event.

mode, is adapted to the link and characteristics.

Figures 2(a) and (b) present the average PSD alongside the extrema for the laboratory and field datasets. To generate these, the data were segmented into 15-minute windows over a 24-hour duration, aligned to the first and third eigenvectors, and smoothed. Notably, the field data exhibits a consistently higher average PSD than the laboratory data by approximately 4 dB within the 0.1–1 Hz range, with a 10–15 dB differential observed between the principal axes. The rotation vector distribution in Figs. 2(c) and (d) demonstrate a higher degree of anisotropy for the ATH transponder compared to the PAL transponder. A comprehensive comparison of anisotropy across all datasets is provided in Fig. 2(e). Although no single dominant factor can fully explain the observed variation in anisotropy, the results consistently point to two contributing sources: terminal-related effects and distributed fiber-induced noise accumulated along the deployed cable routes. Having established the directional noise structure and its anisotropic behavior, we next examine how this knowledge can be leveraged to improve the detection of real seismic events.

Tab. 1: Seismic events under study

Event	Date & Time (UTC)	Magnitude
Event 1	2025-05-22 03:19:35	6,2
Event 2	2025-05-13 22:51:15	6,0
Event 3	2025-07-23 13:26:54	5,1
Event 4	2025-07-07 17:46:04	5,1

We computed the rotation vectors for six transponders to detect four earthquake events (listed in Table. 1) from the Incorporated Research Institutions for Seismology (IRIS) database (M5.1–M6.2 occurred in May and July 2025) [8] within a 40-min window around the event. We compensate for the quasi static drift by using a 10 s running average reference. Eigenvalue decomposition was then applied to another 2 s block-based moving window, which we found to be the optimal size (SNR first increases with the increase in size of this window and then decreases), to align the sig-

nal with the eigenvectors directions and compute the RMS amplitude. The noise floor was estimated from a 10 min pre-event interval (–10 to 0 min). The peak SNR values were also calculated prior to alignment and afterward to assess the impact of anisotropic noise. When reconstructing the continuous amplitude across a rolling window, it is necessary to account for the arbitrary sign ambiguity inherent in eigendecomposition. To prevent artificial phase inversions, phase continuity is enforced by computing the inner product between eigenvectors of successive windows; if the inner product is negative, the sign of the current eigenvector is inverted. Figure 3(b) shows the example of CAT. 1 for event 2 amplitudes where the anisotropy-aware approach significantly improves the best-component SNR of the baseline technique compared the lowest one in Fig. 3(a). The gain comes from noise suppression, not signal enhancement: although the signal is also reduced, the denoising suppresses noise more strongly. Across all transponder–event pairs, the best SNR increases by approximately 2 to 16 dB (Fig. 3(c)). It is also important to note that, in the baseline case, the best SNR is randomly oriented along the default S_1 , S_2 and S_3 directions. In contrast, for the anisotropy-aware approach, it is always aligned with λ_3 , which corresponds to the lowest noise level. Also, the worst SNR in the anisotropy-aware approach is lower than that of the baseline, further highlighting the importance of noise geometry.

Conclusion

This study showed that anisotropy-aware eigenvalue decomposition consistently aligns polarization signals, achieving 2–16 dB SNR improvements across field-deployed subsea transponders, enabling the enhancement of earthquake detection. Future work will focus on expanding the seismic events to broader magnitudes and distances, improving noise characterization by further decoupling terminal-side and fiber-induced contributions, and investigating the benefits for real-time and causal detection operational deployment.

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